Shaila Reddy Kankanala

AI Engineer

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SUMMARY

AI Engineer with a strong focus on Generative AI and Large Language Models, possessing deep expertise in model development, retrieval-augmented generation (RAG), and production-scale machine learning pipelines. Skilled in utilizing frameworks such as TensorFlow and PyTorch, this professional excels in designing and implementing robust AI architectures that significantly enhance model accuracy and reduce inference latency.

PROFESSIONAL EXPERIENCE

Teaching Assistant (AI & Workflows), **San Jose State University | San Jose, CA** **Aug 2025 - May 2026**

- Leveraged OpenAI for sentiment analysis of 300+ student feedback responses, classifying feedback by sentiment and calculating course-level ratings that informed curriculum improvements and feedback implementation.
- Collaborated with 10+ campus departments and Student Wellness Center leadership to translate feedback insights into 25+ initiatives, implementing improvements and reducing response time from 3 days to 1 day.

AI Engineer Intern, **AllyIn.AI | San Jose, CA** **Jul 2025 - Aug 2025**

- Developed PySpark pipelines on GCP to preprocess 250K+ product and user records, leveraging Generative AI techniques with Claude Sonnet for LLM-assisted labeling to generate training data for Llama 3.1 8B fine-tuning.
- Built a LangChain RAG pipeline using Claude Sonnet and Pinecone to process 0.3TB of product-supported PDFs, utilizing Large Language Models for section-based chunking, OpenAI embeddings, metadata filtering, and vector search for Q&A.
- Created Snowflake data models combining product usage, customer activity, and support data for downstream AI and analytics applications, supporting 12+ dashboards and standardized DAU, conversion, and engagement KPIs.
- Implemented dbt and SQL validation checks across 30+ production customer and product data models, validating schemas, nulls, duplicates, and freshness for downstream RAG and analytics applications.

Data Engineer (Applied AI), **Dextara Datamatics | Hyderabad, India** **Aug 2022 - Jun 2024**

- Engineered PySpark and Kafka pipelines integrating sources, preprocessing 2M+ records through cleansing, transformation, deduplication, and validation into training datasets, reducing ingestion latency from 4 hours to 52 minutes.
- Developed AI training-data workflows covering dataset preparation, document preprocessing, labeling, quality validation, and train-validation splits using PySpark and dbt, producing consistent datasets for downstream Machine Learning and Generative AI applications.
- Orchestrated Airflow pipelines for 180+ datasets, automating training-data refreshes, validation, lineage, and delivery to AI workflows, enhancing processing efficiency by 65% and ensuring 99.9% production availability for Machine Learning applications.
- Integrated 20+ REST APIs into cloud data platforms using Docker, Git, and CI/CD automation, enabling scalable ingestion workflows for analytics and machine learning applications.
- Standardized reusable dbt macros, Airflow DAGs, and engineering templates, reducing onboarding time from 21 days to 12 days while improving development consistency and pipeline maintainability.
- Collaborated with product and analytics teams to design Snowflake and Amazon Redshift data models supporting 35+ dashboards, eliminating 15 hours of manual reporting weekly for 50+ business users.

Data Analyst Intern, **RineX.AI | Hyderabad, India** **Jul 2021 - Sep 2021**

- Analyzed reporting data using SQL and Python to identify data quality issues, tracing root causes across ETL workflows and reducing weekly reconciliation effort by 12 hours, ensuring data integrity for AI-driven analytics.
- Developed interactive Tableau and Power BI dashboards tracking 15+ business KPIs, enabling 20+ stakeholders to monitor engagement trends and make faster data-driven decisions.

PUBLICATION

A Generative Model for Stylized Air-Drawing Augmentation-Machine Learning with Applications

Engineered a real-time generative AI pipeline using Python, PyTorch, CUDA, Conv-BiLSTM-Attention, Pix2Pix, Stable Diffusion, and LoRA to transform 225K air-drawn sketches across 45 categories into stylized artwork, achieving 89.75% accuracy, 98.32% Top-3 accuracy, and sub-100ms inference latency.

PROJECTS

AgentOS - Universal Multi-Agent AI Integration Platform

- Built a production-grade agentic AI platform using Python, LangGraph, MCP, A2A, FastAPI, PostgreSQL, Qdrant, Neo4j, Redis, Docker, and Kubernetes with persistent memory, hybrid vector–graph retrieval, multimodal processing.
- Enabling and executing multi-step workflows via a unified integration layer connecting Gmail, Slack, GitHub, Google Drive, Notion, Jira, SQL databases, REST APIs, web search.

MemoSynth-Lite: Multi-Layer RAG Memory Platform

- Built a production-grade Retrieval-Augmented Generation (RAG) platform using Python, LangChain, LlamaIndex, Qdrant, Neo4j, and DuckDB, enabling semantic search, graph reasoning, and analytical querying across structured and unstructured knowledge.
- Developed scalable FastAPI services for memory ingestion, retrieval, summarization, version comparison, and conflict resolution, integrating vector search with graph traversal and Docker-based deployment for modular, production-ready AI applications.

EDUCATION

San Jose State University - *Masters, Applied Data Intelligence* **Aug 2024 - May 2026**

SKILLS

Programming & ML : Python, SQL, PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, CUDA, Conv-BiLSTM, Attention Mechanisms, Model Fine-Tuning, Model Evaluation, Machine Learning, React, AI Technologies

Generative AI & LLMs : Llama 3.1, Claude Sonnet, OpenAI, Stable Diffusion, LoRA, LangChain, LangGraph, LlamaIndex, RAG, Prompt Engineering, Synthetic Data Generation, LLM Evaluation, Agentic AI, MCP, A2A, Large Language Models

AI Data & Retrieval : PySpark, Kafka, dbt, Apache Airflow, Snowflake, Amazon Redshift, Qdrant, Pinecone, Neo4j, DuckDB, Vector Search, Embeddings, Hybrid Retrieval, Graph Retrieval, Metadata Filtering

AI & Data Engineering : ETL/ELT, Data Preprocessing, Data Labeling, Training Dataset Generation, Feature Engineering, Data Validation, Data Quality, Data Lineage, Data Modeling, Distributed Data Processing, Real-Time Streaming

Cloud, Infrastructure & MLOps : GCP, Docker, Kubernetes, FastAPI, Git, CI/CD, ML Pipelines, Model Serving, Inference Optimization, GPU Computing, Production Deployment, Observability

AI Systems & APIs : REST APIs, Multi-Agent Systems, Multi-Step Workflows, Persistent Memory, Multimodal AI, Semantic Search, Graph Reasoning, Contextual Retrieval, Workflow Orchestration, Microservices, OAuth